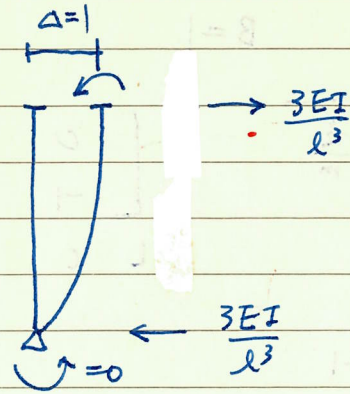
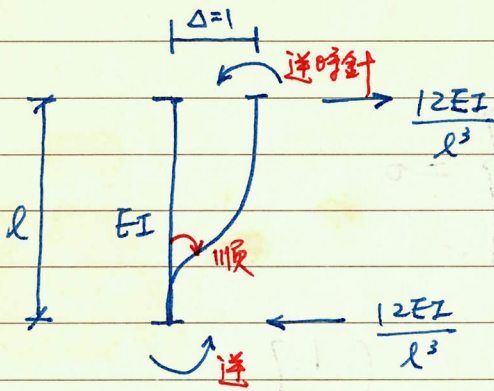


6.

EX. 剪力屋架 求刚度矩阵 [K]

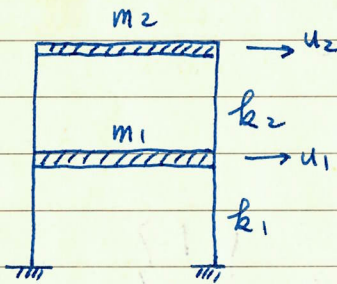


6-6-12-12

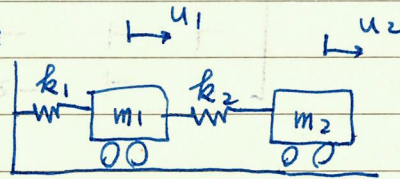
3-0-3-3

弯矩的方向：变形角为顺时针
杆端弯矩与之相反

记住这2个情况，在
结构力学中 就够用了！



转化为数学
model

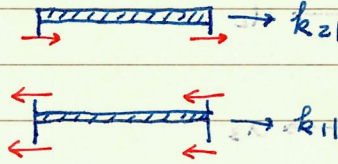
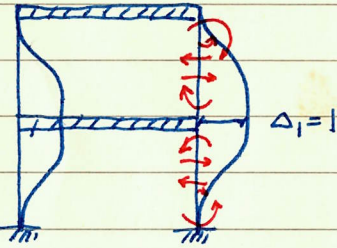


$$[K] = \begin{bmatrix} k_1 + k_2 & -k_2 \\ -k_2 & k_2 \end{bmatrix}$$

记！ (2层楼直接背)

直接取 k 值：

$$\Delta_1 = 1$$



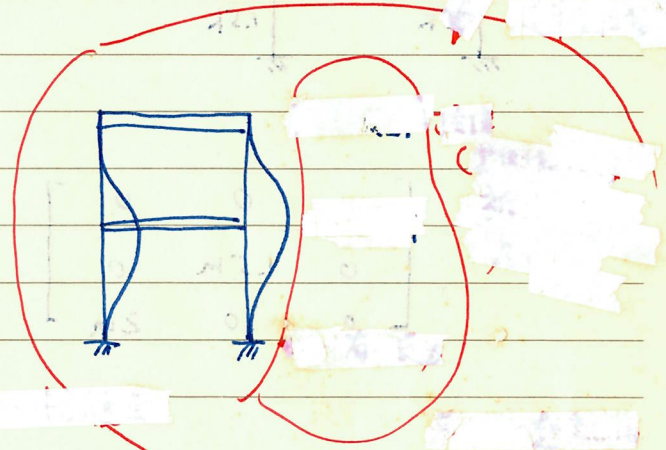
桿端彎矩方向：

由傾角變位決定

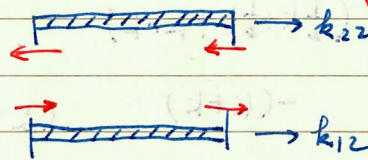
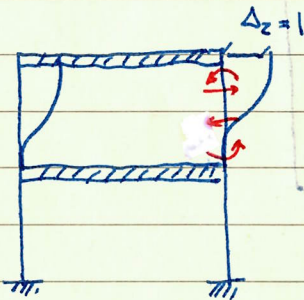
變形角 R 。

順 → 桿端彎矩：逆

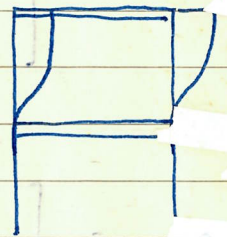
逆 → 桿端彎矩：順



$$\Delta_2 = 1$$



以各杆自由體圖求取
力平衡，



$$[K] = \begin{bmatrix} \Delta_1 = 1 & \Delta_2 = 1 \\ k_{11} & k_{12} \\ k_{21} & k_{22} \end{bmatrix}$$

k_{11} k_{22} ... k_{ii} 為正值

k_{12} k_{21} ... k_{ij} , k_{ji} 為負值

正負號決定後， k 值直接由自由體圖相關之
柱剪力值相加即得。

不必將各樓層之自由體圖畫出，節省時間。